

Prompting Fundamentals on Real Tasks (FL-02)

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Target Task: Summarizing lecture transcripts (From FL-01)

Baseline: Naive Prompt

PROMPT: Summarize this lecture transcript.

OUTPUT EXCERPT: "This lecture was about machine learning. The speaker discussed several topics. In conclusion, it was very informative..."

Notes: The output is incredibly generic. It reads like a book report rather than a useful study tool.

Version 1: + Role Assignment

PROMPT: Act as an expert technical note-taker. Summarize this lecture transcript.

OUTPUT EXCERPT: "Technical Summary: The primary focus was on predictive modeling utilizing Random Forest architectures..."

Change: Added a specific Role Assignment.

Output Difference: The vocabulary immediately became more professional and technical, rather than generic.

Version 2: + Context and Motivation

PROMPT: Act as an expert technical note-taker. I am an intern trying to catch up on a meeting I missed, so I need to know exactly what the next steps are. Summarize this lecture transcript.

OUTPUT EXCERPT: "Technical Summary: ... Next Steps for Intern: You need to review the DuckDB docs."

Change: Added Context and Motivation.

Output Difference: The AI stopped giving a generalized summary and started looking specifically for things that impact my workload.

Version 3: + Output Structure

PROMPT: Act as an expert technical note-taker... Output structure: 1. A 3-sentence executive summary. 2. A bulleted list of 3-5 core concepts. 3. A checklist of action items.

OUTPUT EXCERPT: Executive Summary:

1. ... 2. ... 3. ...

Core Concepts:

- Random Forests

Action Items:

- Review docs

Change: Added strict Output Structure.

Output Difference: The formatting is now perfectly scannable. I don't have to read paragraphs of text anymore.

Version 4: + Few-Shot Examples

PROMPT: ... Output structure: [as above]. Example of a good action item: "- [] Read the DuckDB documentation before tomorrow's sync." Example of a bad action item: "- Learn database stuff."

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OUTPUT EXCERPT: Action Items:

- [] Complete the w05_model.ipynb notebook by Friday 5PM.

Change: Added Few-Shot Examples.

Output Difference: The action items went from vague ("review docs") to highly specific tasks with deadlines attached.

Version 5: + Step Decomposition

PROMPT: ... Step 1: Read the transcript and extract all mentions of tasks or deadlines. Step 2: Extract the core theoretical concepts discussed. Step 3: Format these into the required output structure.

OUTPUT EXCERPT: Step 1 complete. Step 2 complete. Final Output: [Perfectly structured summary]

Change: Added Step Decomposition.

Output Difference: The AI slowed down its reasoning process, resulting in zero missed deadlines from the transcript.

Cross-Model Comparison (Claude vs ChatGPT)

I ran the final Version 5 prompt through both Claude and ChatGPT.

- Accuracy/Structure: Both models followed the step decomposition and output structure perfectly.
- Tone: ChatGPT used slightly more filler words ("In conclusion", "It is important to note"), whereas Claude was much more concise and direct.
- Failure Points: ChatGPT hallucinated one minor deadline that wasn't explicitly stated in the text (inferring it instead), while Claude strictly stuck to the provided text.

Final Reusable Prompt Template

Act as an expert technical note-taker.

Context: I am catching up on a meeting/lecture I missed, and I need to extract actionable next steps.

Step 1: Read the transcript and extract all mentions of tasks, deadlines, or assignments.

Step 2: Extract the core theoretical concepts discussed.

Step 3: Format these into the following structure:

- A 3-sentence executive summary.
- A bulleted list of 3-5 core concepts.
- A checklist of action items.

Examples of good action items: '- [] Review the PR by 5 PM.'

Transcript: [INSERT TRANSCRIPT HERE]